

Uncertainty-Aware Inference at a 13B Scale: How Quantization Affects LLM Confidence Calibration

Team C — Alex Jeon, Chuan Bin Phoe, Samuel Lee, Sanket Makkar
Columbia University
Mentor: Ruchi Mahindru, IBM Research

Abstract—While post-training quantization (PTQ) is computationally effective method to lower LLM inference cost, standard accuracy benchmarks often do not measure *miscalibration*, where a model’s stated confidence in its predictions is misaligned with when it is actually right. In settings where downstream tasks directly utilize probabilities, a miscalibrated answer can have the same amount of negative impact as an incorrect answer. In this work, we measure how five PTQ configurations (FP16, GPTQ-INT8, GPTQ-INT4, AWQ-INT4, NF4) affect calibration on Llama-2 13B across three different Question-Answer (QA) datasets (HellaSwag, TriviaQA, PubMedQA). Furthermore, we compute the Pareto frontiers (across inference cost, task accuracy, and calibration quality) and perform uncertainty based simulation routing studies, comparing the PTQ performances of Llama-2 13B with Llama-1 7B and Mistral-7B¹. We demonstrate that larger language models (LMs) suffer higher miscalibration compared to smaller LMs, and routing is only effective if quantization is faster than the FP16 baseline.

Index Terms—LLM quantization, calibration, ECE, GPTQ, AWQ, NF4, Pareto frontier, uncertainty-based routing

I. INTRODUCTION

Quantization has been proven to be an effective optimization technique to minimize the computational cost and increase the inference speed of an LLM without severe performance degradation [1], [2]. While maintaining high accuracy and low perplexity are key metrics in measuring a model’s performance, it is equally important to measure a model’s calibration and uncertainty. Specifically, even though a quantized model is able to match the baseline performance or better, it may still produce over-confident incorrect predictions and under-confident correct predictions. This can be attributed to the fact that quantization distorts the true output probability distribution [3], [5]. Consequently, understanding this misalignment is crucial in real world applications where the stakes are higher (e.g. medical decision making, financial risk analysis, and autonomous IT operations): *knowing when the model is not certain* about its prediction is as important as the final prediction.

This project systematically studies how PTQ affects LLM confidence calibration across model families and scales. Three teams contributed to this project with Team A (Llama-1 7B), Team B (Mistral-7B), and Team C (Llama-2 13B). However,

this paper covers the contributions of Team C, which can be summarized with the following:

- 1) **Llama-2 13B calibration sweep** using four PTQ methods (GPTQ-INT8/INT4, AWQ-INT4, NF4) with comparisons against a FP16 baseline across hellaswag (commonsense MCQ) [7], pubmedQA (biomedical research QA) [8], and triviaQA (factual QA) [9]. We measure Expected Calibration Error (ECE), Maximum Calibration Error (MCE), and Brier score as well as plot reliability diagrams and per-token entropy.
- 2) **Cross-model calibration analysis** over throughput $\times$ accuracy $\times$ ECE for 15 configurations spanning three model families.
- 3) **Uncertainty-based routing simulator** that sweeps through confidence thresholds to allocate queries between a cheap quantized model and an FP16 escalation path, reporting accuracy, calibration, and cost savings as functions of the threshold.
- 4) **Reproducible analysis pipeline** to run evaluations for the calibration sweep, generate pareto frontiers, and simulate uncertainty-based routing. These scripts can be found in `TeamC/run_eval.py`, `TeamC/pareto_script.py`, `TeamC/routing_simulation.py`.

We make all code and results available on GitHub and Weights & Biases.

II. RELATED WORKS

A. Calibration in neural networks and language models

Confidence calibration measures whether a model’s predicted probabilities align with empirical correctness. Guo et al. [3] established Expected Calibration Error (ECE) and reliability diagrams as standard tools for evaluating miscalibration in modern neural networks, while Naeini et al. [4] introduced adaptive binning approaches that reduce sensitivity to sparse confidence regions. Minderer et al. [15] later revisited calibration in modern architectures and showed that, although calibration has improved under newer training recipes, miscalibration remains a persistent issue.

For language models, calibration is especially important because downstream systems often consume probabilities directly rather than only final predictions. Jiang et al. [6]

¹The respective data are provided by Team A and Team B

studied calibration for language model question answering and showed that strong task performance does not necessarily imply well-calibrated confidence. More broadly, recent work on uncertainty estimation for LLMs has emphasized that model confidence can fail in both overconfident and underconfident regimes, particularly under distribution shift or ambiguous generation settings [5]. Semantic uncertainty methods further argue that token-level confidence alone may miss meaning-preserving alternatives in natural language generation [13].

B. Post-training quantization for LLMs

Post-training quantization (PTQ) has become a standard approach for reducing the memory footprint and inference cost of LLMs without full retraining. GPTQ introduced an accurate Hessian-informed weight-only PTQ procedure for generative transformers and became a widely used baseline for low-bit compression [1]. ZeroQuant likewise demonstrated efficient low-bit quantization for large transformers, highlighting the system-level benefits of compression for deployment [2]. In parallel, Dettmers et al. [12] showed that 8-bit inference can preserve strong model quality at scale, and QLoRA introduced the NF4 datatype as an effective 4-bit representation for quantized language models.

More recent PTQ methods have focused on preserving salient model behavior under aggressive compression. AWQ protects activation-salient weights during quantization and reports strong accuracy-speed trade-offs for LLM inference [11]. These methods are usually evaluated on accuracy, perplexity, or throughput, but such metrics do not directly capture whether the compressed model’s probabilities remain trustworthy. Our work builds on this PTQ literature by treating calibration quality as a first-class deployment metric alongside accuracy and inference cost.

C. Quantization and confidence calibration

The most directly related prior work is Proskurina et al. [5], who studied how GPTQ quantization affects LLM confidence and calibration across several model families. They found that quantization often increases calibration error, decreases confidence in true labels, and disproportionately perturbs examples that were already low-confidence before compression. Their findings provide a strong motivation for examining quantization through the lens of predictive probabilities rather than top-1 accuracy alone.

Our study extends this line of work in three ways. First, instead of limiting the analysis to a single PTQ family, we compare multiple practical quantization configurations used in deployment: GPTQ at INT8 and INT4, AWQ-INT4, and NF4, against an FP16 baseline. Second, we study these effects on *Llama-2 13B* across three question-answering datasets with distinct uncertainty profiles, allowing us to test whether calibration changes are task-dependent. Third, beyond per-model evaluation, we analyze cross-family Pareto trade-offs over throughput, accuracy, and ECE, which connects calibration behavior to systems-level deployment choices.

D. Uncertainty-aware inference and serving trade-offs

A separate but related line of research studies how uncertainty estimates can be used to make inference more efficient. Our results motivates uncertainty-based routing as a deployment strategy in which confident queries are served by a cheaper model and uncertain cases are escalated to a stronger path, turning calibration quality into a systems variable rather than only an evaluation metric. This perspective is closely related to recent work on LLM serving infrastructure, where throughput depends not only on the model and precision but also on the backend implementation.

In particular, vLLM and PagedAttention showed that memory management and serving-stack design can dramatically change practical inference throughput [14]. This is relevant for our setting because a quantized model is only useful as a cheap routing path if its realized serving throughput exceeds that of the full-precision baseline under a comparable stack. Our paper therefore connects model calibration, quantization, and serving efficiency in a unified framework: a quantized model is attractive only when it is not just smaller, but also sufficiently accurate, well-calibrated, and operationally faster.

E. Positioning of our work

Taken together, prior work shows that PTQ can preserve standard performance while still altering predictive confidence in important ways, and calibration remains an under-measured dimension of LLM deployment quality. Existing studies have largely focused either on calibration in full-precision models or on compression quality measured through accuracy and speed. Our work sits at the intersection of these areas by evaluating how practical PTQ choices affect confidence calibration at the 13B scale and by analyzing when those trade-offs remain favorable under uncertainty-aware routing.

III. MODELS AND DATA

TABLE I
TEAM C QUANTIZATION CONFIGURATIONS FOR LLAMA-2 13B.

Config	Method	Bits	Library	Source
FP16	–	16	HF Transformers	meta-llama
GPTQ-INT8	GPTQ	8	AutoGPTQ	TheBloke
GPTQ-INT4	GPTQ	4	AutoGPTQ	TheBloke
AWQ-INT4	AWQ	4	AutoAWQ	TheBloke
NF4	bnb	4	bitsandbytes	on-the-fly

A. Models and quantization configurations

The model family used in this paper is Llama2-13B, which is an auto-regressive language model consisting of 13 billion parameters [10]². We evaluate four PTQ configurations and one FP16 baseline, as shown in Table I. The GPTQ and AWQ models use pre-quantized checkpoints from TheBloke’s Hugging Face repository, whereas NF4 quantization is applied dynamically via `bitsandbytes`. GPTQ is configured with a group size of 128 and activation-ordering enabled.

²Model accessed from HuggingFace via `meta-llama/Llama-2-13b-hf`

B. Datasets

Each of the model configurations (Table I) are evaluated on the hellaswag, triviaqa, and pubmedqa datasets³. These three datasets are selected to cover distinct uncertainty profiles, an essential component for a comprehensive calibration study.

- **HellaSwag**: 4-way commonsense MCQ with 10042 samples designed as reasoning tasks with inherent ambiguity. This dataset tests whether quantization disturbs already-imperfect calibration on confident-correct predictions.
- **PubMedQA**: ternary biomedical research QA with 1000 samples to evaluate an out-of-distribution (OOD) set, which should produce high uncertainty as such questions (and to a larger extent, domain) are not well-represented in the pre-trained data.
- **TriviaQA**: factual QA with multi-token generation with 17944 samples to test if a model expresses low uncertainty on known facts.

For the cross-model and routing analyses, we additionally analyze Team A’s Llama-1 7B and Team B’s Mistral-7B results, with each team’s methodology documented separately.

C. Experiment tracking

Runs are logged to Weights & Biases at this wandb link. Each run is tagged with model, team, quant_method, precision, dataset, seed, and emits tokens_per_second, accuracy, ece as scalar metrics.

Additionally, all diagrams (e.g. entropy, sim-routing, etc...) are saved in each team’s respective results directory (e.g. Team C: https://github.com/chuanbinp/uncertainty-aware-inference/tree/master/TeamC/full_results).

IV. EVALUATION METHODOLOGY

A. Inference and calibration pipeline

All of Team C runs utilizes a single NVIDIA A100 (40GB), HuggingFace Transformers generate() with batch size 1, and deterministic seed 42. The shared evaluation harness (shared/eval_utils.py) implements the calibration pipeline:

- 1) For each example, capture the next-token softmax distribution at the answer position.
- 2) Define *confidence* as $\max_v p(v|\text{context})$ and *accuracy* as the indicator that the argmax matches the gold label.
- 3) Bin examples into 15 equal-width confidence bins, then compute $\text{ECE} = \sum_b \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$, MCE, Brier score, and average per-token entropy.
- 4) Persist the per-example confidence and accuracy tensors as a .pt file for downstream routing.

Each (model, method, precision, dataset) run produces one JSON summary with {model, precision, quant_method, dataset, tokens_per_second, accuracy, ECE, MCE, Brier_Score, Avg_Entropy, num_examples}. This format is consist with the other teams.

³A single seed was used for reproducibility and consistency sake

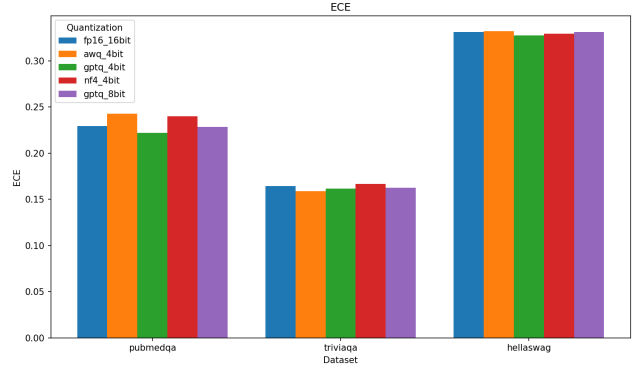


Fig. 1. Llama-2 13B Expected Calibration Error per dataset, grouped by quantization config

B. Cross-model Pareto frontier

Per-model and cross-model Pareto frontiers are generated by comparing inference throughput (tokens per second), task accuracy, and calibration error (ECE). For cross-model analysis, these metrics are averaged over all available results for each (model, quant_method, precision) configuration. We then identify Pareto-optimal configurations that are not dominated by any other configuration in throughput, accuracy, and calibration quality. In addition, we compute dataset-specific and per-model Pareto frontiers to examine how trade-offs differ across datasets and model families.

C. Uncertainty-based routing

We implement a two-tier serving policy: for each example, if the quantized model’s confidence at the answer position exceeds a threshold τ , the prediction is served from the quantized path; otherwise the query switches to FP16. We sweep $\tau \in [0, 1]$ in 101 steps. For each τ :

- **Effective accuracy** is the per-example where-mixture of quantized and FP16 correctness.
- **Effective throughput** is the harmonic mean $(f/T_{\text{quant}} + (1-f)/T_{\text{FP16}})^{-1}$, where f is the fraction of queries served by the quantized path.
- **Cost savings** are reported relative to a uniform FP16 baseline.

For each (model, method, precision, dataset) tuple, we report the τ that maximizes savings subject to the constraint $\text{accuracy}_{\text{routed}} \geq \text{accuracy}_{\text{FP16}} - 0.1\%$. This

is the operating point a deployment engineer would actually pick: “cheapest serving that doesn’t sacrifice accuracy.”

V. EXPERIMENTAL RESULTS

A. Llama-2 13B calibration sweep

Table II reports the full 5×3 matrix of metrics, Fig. 1 plots the ECE column, and Fig. 2 plots the accuracy column.

TABLE II

LLAMA-2 13B CALIBRATION SWEEP: 5 CONFIGURATIONS $\times$ 3 DATASETS. THROUGHPUT (TOK/S) IS HF TRANSFORMERS SINGLE-BATCH GENERATE () ON A SINGLE A100. BOLD ENTRIES HIGHLIGHT THE LOWEST ECE PER DATASET.

Config	HellaSwag				TriviaQA				PubMedQA			
	tok/s	Acc	ECE	Brier	tok/s	Acc	ECE	Brier	tok/s	Acc	ECE	Brier
FP16	1903.394	0.773	0.331	0.269	39.29	0.767	0.164	0.211	527.541	0.470	0.229	0.307
GPTQ-INT8	865.263	0.773	0.331	0.268	18.803	0.767	0.162	0.211	233.754	0.473	0.228	0.308
GPTQ-INT4	2179.729	0.766	0.327	0.269	41.740	0.754	0.162	0.220	660.891	0.472	0.222	0.308
AWQ-INT4	928.439	0.772	0.332	0.274	22.822	0.752	0.165	0.220	288.567	0.470	0.242	0.314
NF4	734.664	0.769	0.329	0.270	19.907	0.757	0.166	0.216	201.293	0.467	0.240	0.390

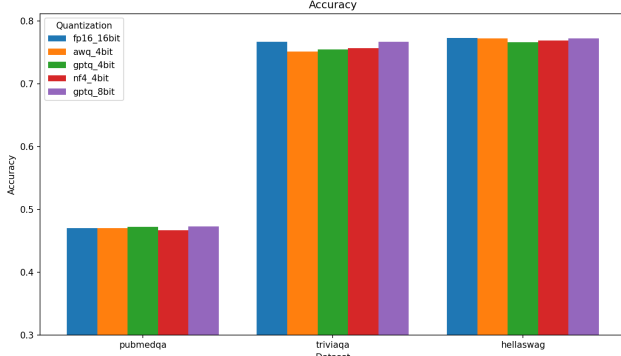


Fig. 2. Llama-2 13B accuracy per dataset, grouped by quantization config

B. Within-model calibration: Llama-2 13B

Finding 1: HellaSwag calibration is largely invariant to quantization at the 13B scale. ECE remains within a narrow band of 0.327 to 0.332 across all five configurations, while accuracy varies by only 0.65% (0.766 to 0.773). The corresponding reliability diagrams are nearly superimposed, indicating minimal calibration drift across quantization settings. These results suggest that 4-bit quantization does not materially affect the calibration or predictive performance of Llama-2 13B on this commonsense multiple-choice completion benchmark.

Finding 2: GPTQ-4bit is the most calibration-preserving 4-bit configuration for Llama-2 13B. Across all three benchmarks, GPTQ-4bit either matches or improves upon the FP16 baseline ECE: on HellaSwag, ECE decreases from 0.331 to 0.327; on TriviaQA, from 0.165 to 0.162; and on PubMedQA, from 0.229 to 0.222. It is also the only 4-bit Llama-2 13B configuration to appear on the within-family Pareto frontier, doing so on TriviaQA and PubMedQA, making it the recommended 4-bit strategy when calibration fidelity is a priority.

Finding 3: AWQ-4bit and NF4 do not reliably preserve calibration on Llama-2 13B. On PubMedQA, both AWQ-4bit (ECE=0.243) and NF4 (ECE=0.240) exceed the FP16 baseline (ECE=0.229), demonstrating that lower-bit quantization can degrade confidence reliability even when accuracy remains similar. NF4 additionally records the highest TriviaQA ECE among all 13B configurations (0.166), above both the FP16

baseline (0.165) and GPTQ-4bit (0.162). Taken together with Finding 2, these results confirm that low-bit quantization does not affect calibration uniformly across methods or tasks within the same model family.

C. Cross-model calibration

Table V ⁴ shows the cross model calibration as well as which models are on the Pareto frontier. Additionally, this GitHub Pages dashboard provides interactive plots for the 2D and 3D Pareto frontiers.

Finding 4: Larger scale does not guarantee better calibration. On HellaSwag, all Llama-2 13B configurations carry the highest ECE values in the study (0.327–0.332), and no Llama-2 13B configuration is Pareto-optimal on that dataset; the only Pareto-optimal quantized configurations are Llama-1 7B GPTQ-4bit (ECE=0.301) and GPTQ-8bit (ECE=0.308), both of which achieve lower calibration error despite having fewer parameters. On TriviaQA and PubMedQA, some Llama-2 13B configurations do reach the Pareto frontier on accuracy–ECE tradeoffs, but they are not cross-model dominant: Mistral-7B and Llama-1 7B configurations occupy distinct frontier segments, demonstrating that scale confers no systematic calibration advantage.

Finding 5: Calibration effects of quantization are dataset- and model-family-specific. AWQ-4bit does not uniformly improve or degrade calibration relative to FP16; its effect reverses direction across datasets and architectures. For Mistral-7B, AWQ-4bit raises ECE on HellaSwag (0.277 to 0.283) and TriviaQA (0.143 to 0.155), yet dramatically lowers ECE on PubMedQA (0.229 to 0.067). For Llama-2 13B, AWQ-4bit reduces ECE on TriviaQA (0.165 to 0.159) but increases it on HellaSwag (0.331 to 0.332) and PubMedQA (0.229 to 0.243). These opposing patterns indicate that quantization-induced confidence shifts are architecture- and task-dependent, cautioning against generalising findings from a single model or benchmark.

Finding 6: Quantization can improve calibration on specific tasks, contrary to the negative trend reported in prior work. On PubMedQA, Mistral-7B AWQ-4bit (ECE=0.067) and GPTQ-4bit (ECE=0.077) both substantially undercut the FP16 baseline (ECE=0.229), representing the largest calibration gains observed in our study; notably, this holds for

⁴Table found within the Appendix

GPTQ despite the general amplification of calibration error reported by Proskurina et al. [5]. However, the improvement does not generalise: on HellaSwag and TriviaQA, Mistral-7B FP16 remains the sole Pareto-optimal configuration for that model family, and all quantized variants increase ECE relative to baseline. These results show that calibration effects are task-contingent and cannot be treated as a universal property of any quantization method.

Finding 7: GPTQ is the most consistently Pareto-efficient post-training quantization family. Across all three evaluation datasets, GPTQ configurations appear on the Pareto frontier more frequently than either AWQ or NF4. On HellaSwag, the only two Pareto-optimal configurations are Llama-1 7B GPTQ-4bit (ECE=0.301, accuracy=73.0%) and GPTQ-8bit (ECE=0.309, accuracy=73.9%). On TriviaQA, GPTQ accounts for four of the eight Pareto-optimal slots: Llama-1 7B GPTQ-4bit (ECE=0.076) and GPTQ-8bit (ECE=0.082), and Llama-2 13B GPTQ-4bit (ECE=0.162) and GPTQ-8bit (ECE=0.162); AWQ appears twice (Llama-1 7B and Llama-2 13B), and FP16 twice (Mistral-7B and Llama-2 13B). On PubMedQA, GPTQ occupies four Pareto slots across three model families: Llama-1 7B GPTQ-4bit and GPTQ-8bit, Mistral-7B GPTQ-4bit, and Llama-2 13B GPTQ-4bit; AWQ is Pareto-optimal for Mistral-7B only, and NF4 does not appear on any frontier across all three datasets. These patterns support a practical recommendation: when the objective is to jointly preserve accuracy and calibration quality under inference-speed constraints, GPTQ is the most reliably Pareto-efficient PTQ family within the scope of this study.

D. Entropy Analysis

Finding 8: Model architecture / QA task has a larger effect compared to that of quantization. Across the 3 model architectures (Llama-2 13B, Llama-1 7B, Mistral-7B), the FP16 and quantized distributions look qualitatively similar for the same benchmarks⁵. This signifies that the main source of variation is not from the reduction of precision but rather the datasets and underlying model architecture. Specifically, quantization preserves the baseline (FP16) model’s uncertainty structure rather than changing it.

Finding 9: While entropy-based separation of correct versus incorrect outputs is task dependent but Mistral generally demonstrates stronger separability when compared to the Llama family. This is especially evident in the HellaSwag dataset, where the separation between the entropy values for correct and incorrect predictions is modest for the Llama models but more apparent for Mistral.

E. Uncertainty-based routing

Finding 10: Confidence-based routing can maintain FP16 accuracy, but the accuracy gains are marginal. Across all three model families, the best accuracy routing threshold usually matches or slightly exceeds FP16 accuracy, but the improvements are small, around 0.07–0.13 percentage

TABLE III
BEST AVG ACCURACY GAIN (VS FP16) FOR EACH MODEL ARCHITECTURE

Architecture	Avg Acc Gain vs FP16
Llama-1 7B	+0.11 PP
Llama-2 13B	+0.13 PP
Mistral 7B	+0.07 PP

TABLE IV
SAVINGS ACROSS LLAMA-1 7B

Config	Dataset	Quant Frac	Acc (vs FP16)	Savings
GPTQ 4-bit	PubmedQA	100%	+0.30pp	69.85%
GPTQ 8-bit	PubmedQA	99.8%	0.00pp	67.78%
GPTQ 8-bit	Hellaswag	87.4%	+0.03pp	59.77%
GPTQ 8-bit	TriviaQA	80.7%	+0.08pp	42.68%
GPTQ 4-bit	Hellaswag	35.5%	+0.03pp	25.07%

point on average (as seen in Table III). Hence, the practical value of routing should be cost reduction under acceptable accuracy and calibration constraints rather than systematically improving predictive accuracy.

Finding 11: Cost savings are not universal but rather concentrated in Llama-1 7B GPTQ. While Llama-1 7B GPTQ models demonstrate positive savings (ranging from 25.07% to 69.85% as referred in Table IV). However, for Llama-2 13B and Mistral-7B, most routing configurations show negative projected savings, implying that routing is only beneficial for specific model / quantization pairing.

F. Limitations and Future Works

While qualitative analysis is done for the entropy distribution analysis between model architectures and across benchmarks, a more quantitative analysis would require additional time and modifications to the codebase. Based on the simulation routing results, the trade-off between a quantized and FP16 path could be affected by the choosing the correct / specific quantized backend (e.g. Marlin for GPTQ-INT4, awq_marlin for AWQ). This would provide a form of standardization w.r.t performance, allowing for more realistic simulation routing comparisons. While this paper focuses on using softmax confidence (e.g. low token confidence as uncertainty), another approach could be to use semantic entropy [13] as the routing signal, especially when considering factual QA could have multiple correct valid surface forms.

VI. CONCLUSION

A. Summary of findings

- 1) **Quantization affects calibration unevenly across model families and tasks.** HellaSwag is quantization-invariant at 13B. On TriviaQA and PubMedQA, GPTQ-INT4 improves Llama-2 13B ECE; AWQ-INT4 and NF4 worsen it on PubMedQA.
- 2) **Pareto leadership is method- and architecture-conditional.** GPTQ-INT4 stays on every per-family frontier we measured. AWQ dominates on Mistral-7B

⁵Please refer to the Entropy Distribution plots in the Appendix.

and is dominated on Llama-2 13B. No Llama-2 13B configuration is cross-model Pareto-optimal.

- 3) **Routing only pays off with matched serving stacks.** Llama-1 7B GPTQ-INT4 returns +69.85% savings on PubMedQA at FP16-equivalent accuracy. Llama-2 13B and Mistral-7B return 0% because FP16-vLLM throughput beats their quantized HF-batched throughput in our setup.

AI TOOL USE DISCLOSURE

AI-assisted tools (including ChatGPT) were used in a limited capacity for proofreading, formatting assistance, and improving clarity of prose after the technical content had already been written by the authors. All experimental design, implementation, profiling, evaluation methodology, analysis, interpretation of results, and conclusions were performed and verified by the authors.

REFERENCES

- [1] E. Frantar, S. Ashkboos, T. Hoefer, and D. Alistarh, “GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers,” in *Proc. ICLR*, 2023.
- [2] Z. Yao, A. Aminabadi, M. Zhang, X. Wu, C. Li, Y. He, and H. Frajnia, “ZeroQuant: Efficient and Affordable Post-Training Quantization for Large-Scale Transformers,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [3] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On Calibration of Modern Neural Networks,” in *Proc. ICML*, 2017.
- [4] M. P. Naeini, G. F. Cooper, and M. Hauskrecht, “Obtaining Well Calibrated Probabilities Using Bayesian Binning into Quantiles,” in *Proc. AAAI*, 2015.
- [5] I. Proskurina, L. Galke, F. Sileo, and J. Velcin, “When Quantization Affects Confidence of Large Language Models?,” in *Findings of NAACL*, 2024.
- [6] Z. Jiang, J. Araki, H. Ding, and G. Neubig, “How Can We Know When Language Models Know? On the Calibration of Language Models for Question Answering,” in *Transactions of the Association for Computational Linguistics*, 2021.
- [7] R. Zellers, A. Holtzman, H. Bisk, A. Farhadi, and Y. Choi, “HellaSwag: Can a Machine Really Finish Your Sentence?,” in *Proc. ACL*, 2019.
- [8] Q. Jin, B. Dhingra, Z. Liu, W. W. Cohen, and X. Lu, “PubMedQA: A Dataset for Biomedical Research Question Answering,” in *Proc. EMNLP-IJCNLP*, 2019.
- [9] M. Joshi, E. Choi, D. Weld, and L. Zettlemoyer, “TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension,” in *Proc. ACL*, 2017.
- [10] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, et al., “Llama 2: Open Foundation and Fine-Tuned Chat Models,” arXiv preprint arXiv:2307.09288, 2023.
- [11] J. Lin, J. Tang, H. Tang, S. Yang, X. Dang, and S. Han, “AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration,” in *Proc. MLSys*, 2024.
- [12] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient Finetuning of Quantized LLMs,” in *Proc. NeurIPS*, 2023.
- [13] L. Kuhn, Y. Gal, and S. Farquhar, “Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in NLG,” in *Proc. ICLR*, 2023.
- [14] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, and I. Stoica, “Efficient Memory Management for Large Language Model Serving with PagedAttention,” in *Proc. SOSP*, 2023.
- [15] M. Minderer et al., “Revisiting the Calibration of Modern Neural Networks,” in *Proc. NeurIPS*, 2021.

APPENDIX

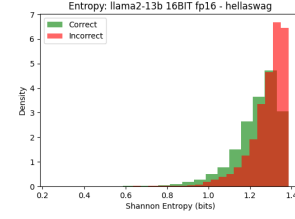


Fig. 3. Llama-2 13B FP16: Entropy Distribution for HellaSwag

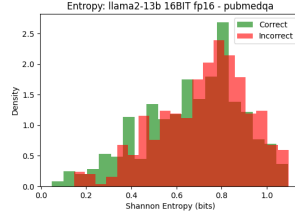


Fig. 4. Llama-2 13B FP16: Entropy Distribution for PubMedQA

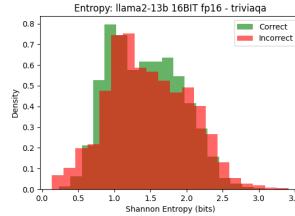


Fig. 5. Llama-2 13B FP16: Entropy Distribution for TriviaQA

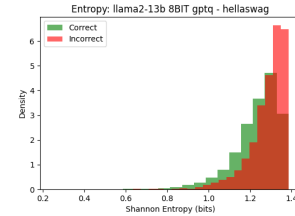


Fig. 6. Llama-2 13B 8-bit GPTQ: Entropy Distribution for HellaSwag

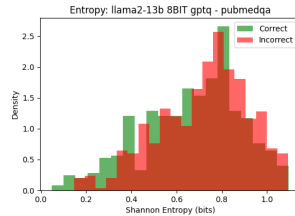


Fig. 7. Llama-2 13B 8-bit GPTQ: Entropy Distribution for PubMedQA

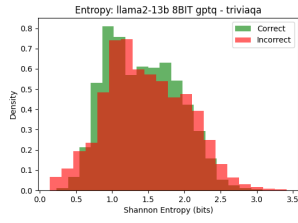


Fig. 8. Llama-2 13B 8-bit GPTQ: Entropy Distribution for TriviaQA

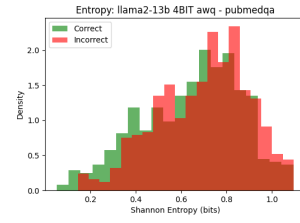


Fig. 13. Llama-2 13B AWQ: Entropy Distribution for PubmedQA

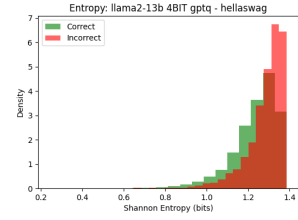


Fig. 9. Llama-2 13B 4-bit GPTQ: Entropy Distribution for Hellaswag

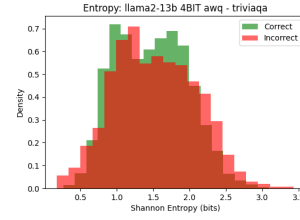


Fig. 14. Llama-2 13B AWQ: Entropy Distribution for TriviaQA

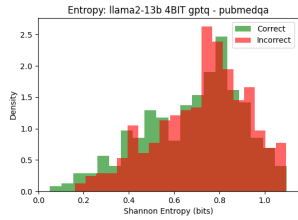


Fig. 10. Llama-2 13B 4-bit GPTQ: Entropy Distribution for PubmedQA

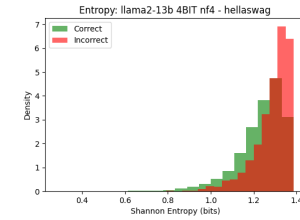


Fig. 15. Llama-2 13B NF4: Entropy Distribution for HellaSwag

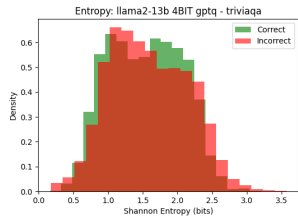


Fig. 11. Llama-2 13B 4-bit GPTQ: Entropy Distribution for TriviaQA

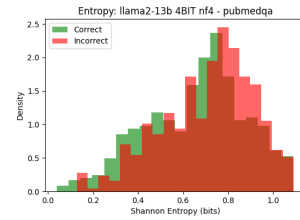


Fig. 16. Llama-2 13B NF4: Entropy Distribution for PubmedQA

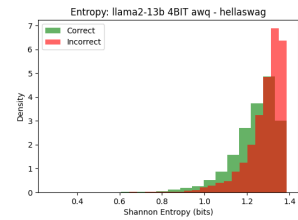


Fig. 12. Llama-2 13B AWQ: Entropy Distribution for Hellaswag

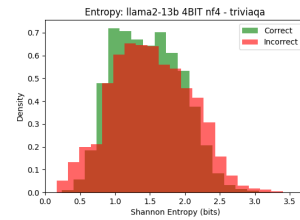


Fig. 17. Llama-2 13B NF4: Entropy Distribution for TriviaQA

TABLE V
BENCHMARK RESULTS ACROSS DATASETS AND QUANTIZATION
CONFIGURATIONS.

Dataset	Model	Precision	Quant	Tokens/s	Accuracy	ECE	Pareto
HellaSwag	LLaMA-1-7B	4bit	AWQ	671.8	0.7328	0.3040	
HellaSwag	LLaMA-1-7B	4bit	GPTQ	4181.3	0.7300	0.3015	✓
HellaSwag	LLaMA-1-7B	4bit	NF4	726.7	0.7342	0.3021	
HellaSwag	LLaMA-1-7B	8bit	GPTQ	3883.1	0.7394	0.3085	✓
HellaSwag	LLaMA-1-7B	16bit	FP16	1226.7	0.7395	0.3086	
HellaSwag	LLaMA-2-13B	4bit	AWQ	928.4	0.7721	0.3320	
HellaSwag	LLaMA-2-13B	4bit	GPTQ	2179.7	0.7664	0.3273	
HellaSwag	LLaMA-2-13B	4bit	NF4	734.7	0.7690	0.3294	
HellaSwag	LLaMA-2-13B	8bit	GPTQ	865.3	0.7726	0.3309	
HellaSwag	LLaMA-2-13B	16bit	FP16	1903.4	0.7729	0.3313	
HellaSwag	Mistral-7B	4bit	AWQ	1221.3	0.8192	0.2825	
HellaSwag	Mistral-7B	4bit	GPTQ	1002.4	0.8246	0.2893	
HellaSwag	Mistral-7B	4bit	NF4	788.6	0.8259	0.2793	
HellaSwag	Mistral-7B	8bit	GPTQ	1007.6	0.8191	0.2827	
HellaSwag	Mistral-7B	16bit	FP16	2224.9	0.8293	0.2771	✓
PubMedQA	LLaMA-1-7B	4bit	AWQ	349.5	0.5390	0.2552	
PubMedQA	LLaMA-1-7B	4bit	GPTQ	1292.3	0.5390	0.2402	✓
PubMedQA	LLaMA-1-7B	4bit	NF4	221.5	0.5320	0.2469	
PubMedQA	LLaMA-1-7B	8bit	GPTQ	1214.4	0.5350	0.2310	✓
PubMedQA	LLaMA-1-7B	16bit	FP16	389.6	0.5360	0.2392	✓
PubMedQA	LLaMA-2-13B	4bit	AWQ	288.6	0.4700	0.2426	
PubMedQA	LLaMA-2-13B	4bit	GPTQ	660.9	0.4720	0.2218	✓
PubMedQA	LLaMA-2-13B	4bit	NF4	201.3	0.4670	0.2400	
PubMedQA	LLaMA-2-13B	8bit	GPTQ	233.8	0.4730	0.2284	
PubMedQA	LLaMA-2-13B	16bit	FP16	527.5	0.4700	0.2293	
PubMedQA	Mistral-7B	4bit	AWQ	335.0	0.4920	0.0667	✓
PubMedQA	Mistral-7B	4bit	GPTQ	276.9	0.4930	0.0766	✓
PubMedQA	Mistral-7B	4bit	NF4	213.0	0.4810	0.2361	
PubMedQA	Mistral-7B	8bit	GPTQ	272.5	0.4610	0.1033	
PubMedQA	Mistral-7B	16bit	FP16	621.7	0.4910	0.2293	✓
TriviaQA	LLaMA-1-7B	4bit	AWQ	28.8	0.6752	0.0707	✓
TriviaQA	LLaMA-1-7B	4bit	GPTQ	73.9	0.6494	0.0763	✓
TriviaQA	LLaMA-1-7B	4bit	NF4	25.7	0.6616	0.0750	
TriviaQA	LLaMA-1-7B	8bit	GPTQ	68.1	0.6860	0.0823	✓
TriviaQA	LLaMA-1-7B	16bit	FP16	32.1	0.6854	0.0837	
TriviaQA	LLaMA-2-13B	4bit	AWQ	22.8	0.7514	0.1586	✓
TriviaQA	LLaMA-2-13B	4bit	GPTQ	41.7	0.7545	0.1617	✓
TriviaQA	LLaMA-2-13B	4bit	NF4	19.9	0.7569	0.1664	
TriviaQA	LLaMA-2-13B	8bit	GPTQ	18.8	0.7672	0.1625	✓
TriviaQA	LLaMA-2-13B	16bit	FP16	39.3	0.7671	0.1645	✓
TriviaQA	Mistral-7B	4bit	AWQ	31.3	0.6854	0.1545	
TriviaQA	Mistral-7B	4bit	GPTQ	24.6	0.6807	0.1525	
TriviaQA	Mistral-7B	4bit	NF4	23.9	0.6870	0.1581	
TriviaQA	Mistral-7B	8bit	GPTQ	24.1	0.6778	0.1550	
TriviaQA	Mistral-7B	16bit	FP16	51.8	0.7005	0.1427	✓